The faster `f4_fp` in the end-to-end index calculus: 5.0× cheaper, same relations

2026-10-18. Run report: `reports/run-20261018T1548Z/report.pdf`. Engine change: PR #1717 (`research/f4_fp_msolve_parity_20261011`).

Question

#1717 made `f4_fp` level with msolve on generic $\mathbb{F}_p$ systems. Which index-calculus pipelines call it, and what does the change do to their end-to-end cost?

Where `f4_fp` runs

The Koblitz binary-field pipeline does not use `f4_fp`; it runs the Boolean engines (inherited F4, F6-IC, `gf2_elim`). `f4_fp` is the decomposition solver of the prime-field and cover pipelines: `jv_cover`, `jv_quartic`, `jv_quintic`, `jv_sieve`, `glv_gaudry`, `coordinate_descent`, `sig_fp_tower`, and the P-256 factor-base binaries. This study measures `jv_cover`, an index calculus on $E(\mathbb{F}_{p^6})$ through a genus-2 cover with Nagao decompositions, end to end.

Result

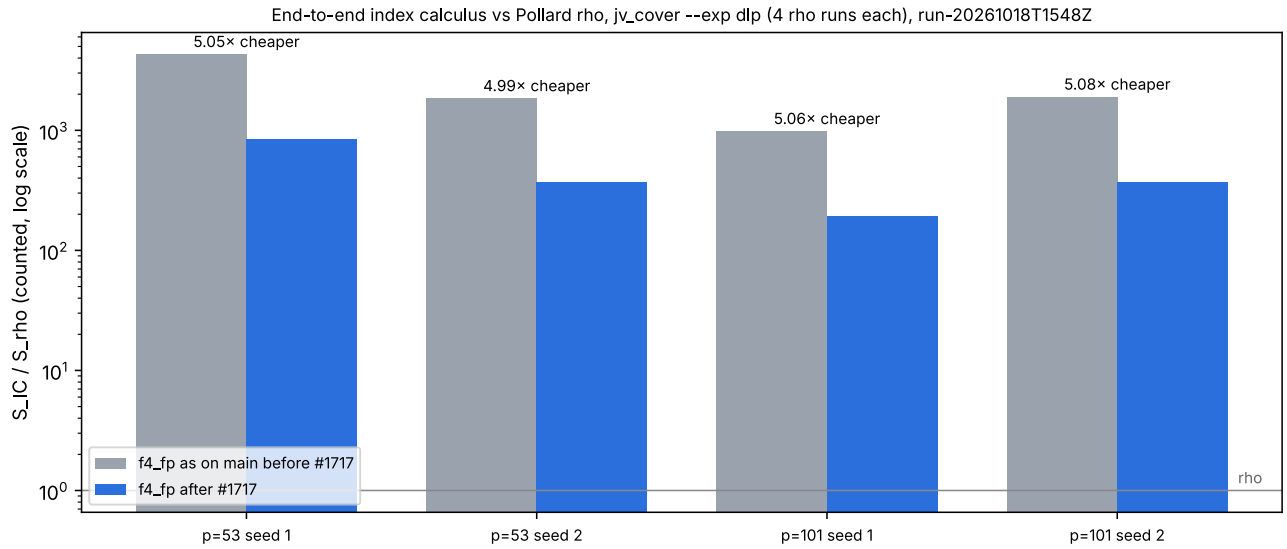


Figure 1: End-to-end $S_{\mathrm{IC}}/S_{\mathrm{rho}}$, before and after #1717, run-20261018T1548Z.

p, seed	S before ÷ after	S/S_{rho} before	S/S_{rho} after	answer
53, 1	5.05	4297	850	correct, same relations
53, 2	4.99	1843	370	correct, same relations
101, 1	5.06	981	194	correct, same relations
101, 2	5.08	1897	373	correct, same relations

Ratios derived from the recorded S; the class is **engineering** (same algorithm, cheaper engine).

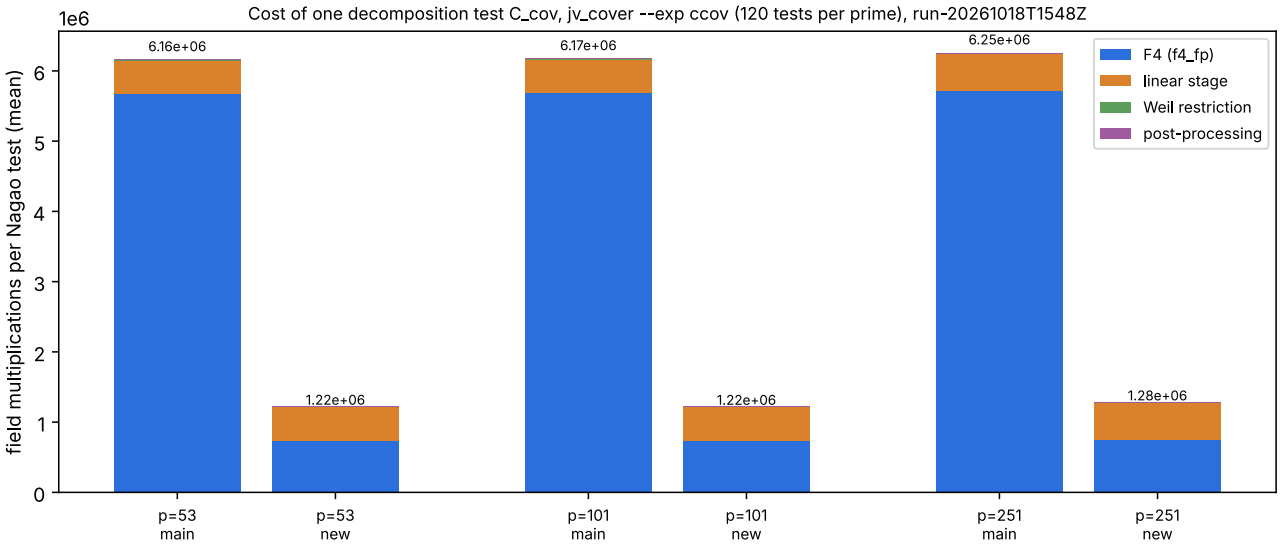


Figure 2: Cost of one decomposition test by stage.

Per decomposition test, the F4 stage costs 7.7× fewer field multiplications (5.7e6 ⟶ 7.4e5) and C_{cov} 5.0× fewer (6.2e6 ⟶ 1.2e6); the reduction is the same at p = 53, 101 and 251. Gebauer–Möller drops the degree-9 and degree-10 steps that only reduced redundant pairs: F4 now stops at degree 8, with 231 rows instead of 767. The linear stage after F4 is unchanged (4.8e5) and is now 40 % of a test, so it is the next lever in this pipeline.

Distance to rho, scope and limits

- **Distance to rho.** After the change this pipeline still costs 194–850× the matched Pollard rho at these sizes (4 rho runs each, so the reference is noisy). That is the distance remaining.
- **Lower-bound comparison.** Neither arm emits counts for its unpriced native work, so the §5a band cannot be applied: S/S_{rho} is a lower-bound comparison, not a ranking.
- **What is measured.** Counted cost, deterministic and identical across hosts. Wall time (912 s ⟶ 370 s for the whole dlp invocation, rho runs included) is shown as an observation only; the host is shared (isolation L0).
- **Scope.** p = 53 and 101 end to end, p up to 251 per test, one run each. The engine change is the only difference between the arms.
- **Leaderboard.** Updated in this PR with its own component section, outside tables A–C, read from `summary.json`.